Fiscal Year:	FY 2005	**Task Last Updated:**	FY 05/24/2005
PI Name:	Derby, Jeffrey J. Ph.D.		
Project Title:	Quantitative Models for Microgravity Melt Crystal Growth with Additional Applications to Detached Solidification and II-VI Materials		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Basics of solidification		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	derby@umn.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	612-625-8881
Organization Name:	University of Minnesota		
PI Address 1:	Department of Chemical Engineering and Materials Science		
PI Address 2:	151 Amundson Hall, 421 Washington Avenue, SE		
PI Web Page:			
City:	Minneapolis	**State:**	MN
Zip Code:	55455-0132	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	11/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Su, Ching-Hua	**Contact Phone:**	256-544-7776
Contact Email:	ching.h.su@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNM04AA19G		
Performance Goal No.:			
Performance Goal Text:			

Large single-crystals of exacting quality are needed in the fabrication of advanced optical, magnetic, and electronic devices, and the growth of these crystals presents some of the most difficult challenges today in materials processing. Crystal growth conducted under microgravity conditions has had a profound impact on improving our understanding of these systems, and the research proposed here aims to further this knowledge. The ground-based research program undertaken here will develop powerful, new computational models for the analysis and optimization of microgravity crystal growth processes. These models will couple a realistic and user-friendly model for furnace heat transfer with a powerful finite element code for computing continuum transport and interfacial phenomena in crystal growth. Parallel with this work, these models will be applied to study phenomena of great significance in melt crystal growth, specifically three-dimensional, transient effects on melt convection and segregation during the growth of II-VI materials. A material of particular interest is cadmium zinc telluride, an important strategic material used in portable, high-performance

Task Description:	radiation detectors. In addition to its microgravity relevance, this work promises many other significant benefits to ground-based research and production of crystals grown from the melt. In general terms, the understanding gained from successful modeling of crystal growth systems will lead to better process operation and design, ultimately yielding better quality crystals at higher production rates and lower costs. In addition, the computational challenges of realistically modeling these systems are significant, and the continued development of algorithms for high performance computing will be noteworthy in itself. This work will also feature significant international collaboration, outreach, and academic achievement. This research will be conducted with international partners, promising a symbiosis of cultural and research approaches that will benefit both partners and nations in the areas of microgravity and materials science research and education. Finally, the multi-disciplinary and technologically relevant nature of this work promises to impart the graduate students involved with modern skills needed to advance materials processing research and practice of tomorrow.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Over the first year of the project, we have addressed two important issues toward our goal of a quantitatively predictive model for melt crystal growth. First, we have successfully coupled our finite element codes with a sophisticated code for modeling global furnace heat transfer, namely the CrysVUN code of Mueller et al. Second, we have successfully implemented a high-dimensional basis set for computing three-dimensional, incompressible melt flows. Both of these items are described in more detail in the ensuing paragraphs. Coupling a global heat transfer code with our finite element crystal growth codes has proven to be more challenging than anticipated because of several mathematical and computational issues arising from model coupling. However, we have successfully developed a novel, two-parameter, loose coupling scheme to exchange temperature and heat flux information between the two models. The framework of fixed-point iterations has been employed to assess the convergence behavior of this modular, block Gauss--Seidel iteration of the independent models as a function of the coupling parameters. Certain choices of coupling parameters that are physically reasonable were shown to result in algorithms that will never converge, demonstrating that notions based on physical intuition may not be useful for predicting algorithm performance. We are currently developing approximate-Newton algorithms using strongly coupled iterations for our crystal growth model. These methods promise convergence rates approaching quadratic and far greater robustness compared to the loosely coupled, block Gauss--Seidel approach. The second accomplishment is motivated by the need of a more robust and accurate computational technique to solve for three-dimensional, steady-state, incompressible flows representative of those found in melt crystal growth systems. Traditional approaches have employed equal-order interpolation using linear basis functions on tetrahedral elements using a pressure-stabilized, Petrov-Galerkin method (PSPG). Our new approach uses a Galerkin mixed-order formulation employing a higher-order, quadratic representation of the velocity field with a linear basis for pressure. This basis set is a modification of the Crouzeix--Raviart element, containing 15 nodes, 45 velocity degrees of freedom, and 4 pressure degrees of freedom per tetrahedral element. Both algorithms employ Newton's method with an iterative linear solver and are implemented using MPI protocols on parallel computers. Several preconditioning methods are employed with GMRES via PETSc for the linear solver. Ongoing work will compare the accuracy and robustness of these methods.
Bibliography Type:	Description: (Last Updated: 09/04/2025)
Articles in Peer-reviewed Journals	Yeckel A, Derby JJ. "Dynamics of three-dimensional convection in microgravity crystal growth: g-jitter with steady magnetic fields." Journal of Crystal Growth, 2004 Mar;263(1-4):40-52. http://dx.doi.org/10.1016/j.jcrysgro.2003.11.047 , Mar-2004
Articles in Peer-reviewed Journals	Yeckel A, Compere G, Pandy A, Derby JJ. "Three-dimensional imperfections in a model vertical Bridgman growth system for cadmium zinc telluride." Journal of Crystal Growth, 2004 Mar;263(1-4)629-44. http://dx.doi.org/10.1016/j.jcrysgro.2003.12.025 , Mar-2004
Books/Book Chapters	Derby JJ, Yeckel A. "Modeling of Crystal Growth Processes." in "Crystal Growth -- From Fundamentals to Technology." Ed. G. Muller, J.J. Metois, P. Rudolph. Amsterdam : Elsevier, 2004. p. 143-167., Jul-2004
Dissertations and Theses	Pandy A. "Global modeling of bulk crystal growth of cadmium zinc telluride in industrial VB/VGF systems." Dissertation, University of Minnesota, July 2004. , Jul-2004
Presentation	Pandy, A.; Derby, J.J. "Modeling an industrial vertical Bridgman process for CZT growth: Coupling furnace heat transfer and crystal growth models" AACGE/West, American Association for Crystal Growth Western Regional Section Conference on Crystal Growth and Epitaxy, Fallen Leaf Lake, CA Jun-2004
Presentation	Sonda, P.; Jung, T.; Kellner, M.; Hadler, H.; Treibig, J. ; Mueller, G.; Yeckel, A,; Derby, J.J. "The Global Three-dimensional Simulation of the Vertical Bridgman Process" ICCG-14, The Fourteenth International Conference on Crystal Growth, Grenoble, France Aug-2004
Presentation	Yeckel, A.; Compere, G.; Pandy, A.; Derby, J.J. "Three-dimensional imperfections in a model vertical Bridgman growth system for cadmium zinc telluride" ICCG-14, The Fourteenth International Conference on Crystal Growth, Grenoble, France Aug-2004
Presentation	Derby, J.J. "Modeling of Crystal Growth Processes" ISSCG-12, The Twelfth International Summer School on Crystal Growth, Berlin, Germany Aug-2004

Presentation	Derby, J.J. "Computing moving boundaries in large-scale simulations of melt crystal growth" Sixth World Congress on Computational Mechanics (WCCM VI), Beijing, China Sep-2004
Presentation	Derby, J.J. "A Broader Overview of the Role of Modeling in the Development of Growth Methods for CdZnTe" The 2004 U.S. Workshop on the Physics and Chemistry of II-VI Materials, Chicago, IL Oct-2004
Presentation	Derby, J.J. "Multi-scale modeling of cadmium zinc telluride growth in an industrial crystal growth system" American Institute for Chemical Engineering Annual Meeting, Austin, TX Nov-2004